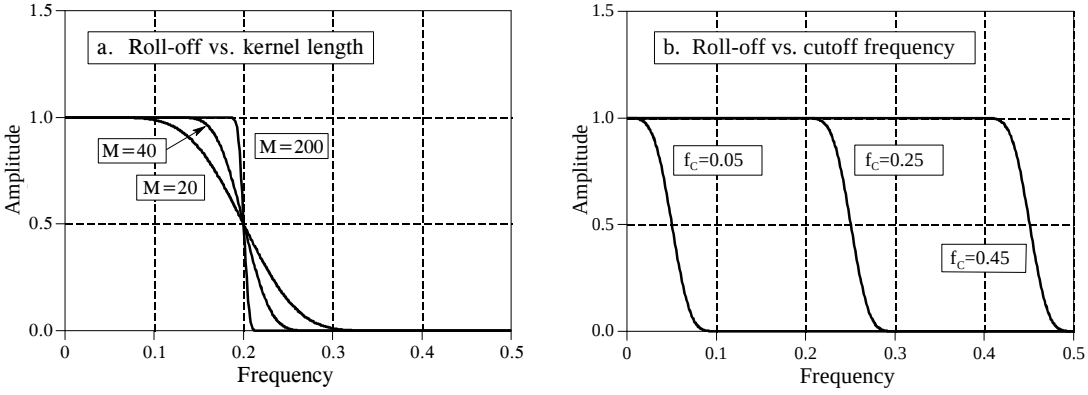

FIGURE 16-3
Filter length vs. roll-off of the windowed-sinc filter. As shown in (a), for M = 20, 40, and 200, the transition bandwidths are BW = 0.2, 0.1, and 0.02 of the sampling rate, respectively. As shown in (b), the shape of the frequency response does not change with different cutoff frequencies. In (b), M = 60.

is expressed as a fraction of the sampling rate, and therefore must be between 0 and 0.5. The value for M sets the *roll-off* according to the approximation:

EQUATION 16-3
Filter length vs. roll-off. The length of the filter kernel, M, determines the transition bandwidth of the filter, BW. This is only an approximation since roll-off depends on the particular window being used.

$$M \approx \frac{4}{BW}$$

where BW is the width of the transition band, measured from where the curve just barely leaves one, to where it almost reaches zero (say, 99% to 1% of the curve). The transition bandwidth is also expressed as a fraction of the sampling frequency, and must between 0 and 0.5. Figure 16-3a shows an example of how this approximation is used. The three curves shown are generated from filter kernels with: M = 20, 40, and 200. From Eq. 16-3, the transition bandwidths are: BW = 0.2, 0.1, and 0.02, respectively. Figure (b) shows that the shape of the frequency response does not depend on the cutoff frequency selected.

Since the time required for a convolution is proportional to the length of the signals, Eq. 16-3 expresses a trade-off between *computation time* (depends on the value of M) and *filter sharpness* (the value of BW). For instance, the 20% slower roll-off of the Blackman window (as compared with the Hamming) can be compensated for by using a filter kernel 20% longer. In other words, it could be said that the Blackman window is 20% slower to execute that an equivalent roll-off Hamming window. This is important because the execution speed of windowed-sinc filters is already terribly slow.

As also shown in Fig. 16-3b, the cutoff frequency of the windowed-sinc filter is measured at the *one-half amplitude* point. Why use 0.5 instead of the